$\Lambda \to p\pi^-$ in the ePIC B0 tracker: acceptance, tracking efficiency and the reconstructed Λ mass

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Abstract

With a single-Λ particle gun into the far-forward region of ePIC (p = 50–100 GeV, θ = 8–18 mrad, 25 mrad crossing angle), we study the reconstruction of $\Lambda \to p\pi^-$ in the B0 tracker, for the B0 geometry before and after eic/epic#1194. The geometric acceptance along the Λ direction drops from 95 % to 70 % in the new geometry, which has a gap on the electron-beam side. Both Λ daughters cross at least three B0 layers in 63 % (old) and 41 % (new) of the $p\pi^-$ decays. With truth seeding in EICrecon main, the tracking efficiency for accepted daughters is 82–93 % for the proton and 56–61 % for the π^-; the π^- is lost because it is soft, in the track finder. The thin (50 μm) sensors of the new geometry need a 5 keV threshold instead of the default 10 keV. Altogether 15 % (old), 10 % (new, 10 keV) and 12 % (new, 5 keV) of the generated Λs have both daughters reconstructed. The reconstructed mass peaks at the PDG value with $\sigma \approx$ 6–7 MeV, and a single cut from the two-body kinematics, $p(+)/p(-) > 2.5$, selects the signal with $\approx$ 97 % purity without using truth information.

Contents

1 Introduction and motivation **3**

2 Setup **3**
2.1 Software . . . 3
2.2 Samples . . . 3
2.3 Sample composition . . . 3

3 Definitions **4**

4 First look: B0 hits and the missing tracks **4**

5 Geometric acceptance along the Λ direction **5**

6 Acceptance for the Λ daughters **6**
6.1 Decay kinematics . . . 6
6.2 Results . . . 7

7 Tracking efficiency **9**
7.1 The B0 threshold . . . 10

8 Acceptance $\times$ efficiency **11**

9 The Λ mass **12**
9.1 Truth-matched pairs . . . 12
9.2 Without truth in the selection . . . 12

10 Summary and outlook **14**

A Reproducing the results **14**

B Pitfalls met **14**

1 Introduction and motivation

Far-forward Λ hyperons are central to several EIC measurements, e.g. Λ production and polarization in the target fragmentation region, and Λ tagging in exclusive and diffractive reactions. At the top energies, a large fraction of these Λs fly at a few to ~ 20 mrad from the proton beam and decay metres downstream of the interaction point (IP). With $c\tau = 7.89$ cm, a 75 GeV Λ has a mean decay length of ≈ 5.3 m, comparable to the distance of the B0 tracker ($z \approx 5.8$–6.9 m). The charged decay $\Lambda \to p\pi^-$ (64 %) must then be reconstructed in the B0 tracker, inside the B0 dipole magnet, from tracks that do not come from the IP.

The goal of this work was to reconstruct the Λ mass from B0 tracks step by step, and to understand each loss along the way: what fraction of decays the detector covers (acceptance), what fraction of the covered daughters are reconstructed (tracking efficiency), and the product. The study also compares the B0 geometry of the eic-shell nightly at the time with the realistic B0 geometry of eic/epic#1194, and the EICrecon of the nightly with EICrecon main, which contains the B0 truth seeding for displaced vertices of EICrecon#2901.

2 Setup

2.1 Software

Simulation	`ddsim` (DD4hep/Geant4) inside eic-shell (`eicweb/eic_xl:nightly`), `Geant4TVUserParticleHandler`
Old geometry	epic `43f427e` (nightly of 2026-11-02): B0 sensitive layers 300 μm
New geometry	epic main `ec774d8` = merge of eic/epic#1194 (realistic B0): sensors 50 μm
Reconstruction	EICrecon main `8b44cae` (local build), contains EICrecon#2901
	EICrecon `a7bab3f` of the nightly used only in Sec. 4
Tracking	ACTS CKF, *truth seeding* (`B0TrackerTruthSeeds`)
Analysis	ROOT 6.34 (LCG_107), compiled macros, this repository

2.2 Samples

All samples use the same particle gun: Λ with p uniform in 50–100 GeV, θ uniform in 8–18 mrad and ϕ uniform, in the 18×275 GeV craterlake geometry, with `-crossingAngleBoost -0.025` (25 mrad crossing angle; the gun angles are w.r.t. the proton beam).

- 100 events per geometry (seed 42): first look, comparison of EICrecon versions.
- 1000 events per geometry (seeds 101 and 102, 500 events each): all main results. Each is reconstructed with EICrecon main with the default B0 threshold (10 keV) and with 5 keV.
- 5000 geantinos per geometry at 100 GeV with the same angles: geometric acceptance.

The ROOT files are on EOS (`/eos/user/g/gninio/eic/`); the macros and scripts are in the repository gadininio/eic-B0-detector, and `scripts/make_results.sh` produces every plot and number of this report.

2.3 Sample composition

Of the generated Λs, 48–50 % decay to $p\pi^-$, 25–28 % to $n\pi^0$, and 24–25 % end otherwise: they interact in material (beam pipe, B0 region) or leave the world volume before decaying. 88–90 % of the $p\pi^-$ decays happen before the B0 tracker ($z < 5.8$ m). The invariant mass of the true p and π^- momenta is exactly 1.115683 GeV for all $p\pi^-$ decays (`lambda_mass_truth.C`).

3 Definitions

$\Lambda \to p\pi^-$ decay. A generated Λ (generatorStatus 1) with exactly two daughters, a proton and a π^-. A Λ that interacts in material is also the parent of the interaction products, but has more than two daughters; without this condition, pions from such interactions were counted as Λ daughters.

B0 layers. Hit z positions are grouped into layers: a gap larger than 50 mm starts a new layer (the four layers are 270–420 mm apart). A particle needs hits in at least three layers for a track.

Angles. θ, ϕ, θ_x, θ_y are given w.r.t. the proton beam, i.e. after undoing the 25 mrad rotation of the crossing-angle boost.

Geometric acceptance (along the Λ direction). Fraction of straight lines from the IP at the Λ gun angles that cross ≥ 3 B0 layers, from a geantino gun (no charge, no interactions). This is pure detector coverage, *not* an acceptance for Λs: the Λ is neutral and leaves no hits.

Daughter acceptance. For each $p\pi^-$ decay: is the daughter's own set of simulated B0 hits (Geant4, before digitization) spread over ≥ 3 layers? Given for the proton, the π^-, and both daughters of the same Λ:

$$A = \frac{\text{daughters with simulated hits in} \geq 3 \text{ layers}}{\Lambda \to p\pi^- \text{ decays}}.$$

It includes the decay position, the opening angle, the B0 dipole field and material, but no detector response.

Reconstructed daughter. A `ReconstructedTruthSeededChargedParticle` whose best truth match (highest weight in `ReconstructedTruthSeededChargedParticleAssociations`) is the daughter.

Tracking efficiency.

$$\varepsilon = \frac{\text{accepted daughters that are reconstructed}}{\text{accepted daughters}}.$$

The geometry cancels; it includes digitization (threshold), seeding, track finding (CKF), track filtering and particle building.

Acceptance $\times$ efficiency.

$$A \times \varepsilon = \frac{\text{reconstructed daughters}}{\Lambda \to p\pi^- \text{ decays}},$$

same denominator as the acceptance. It factorizes into $A \cdot \varepsilon$ up to daughters that are reconstructed without being accepted (at most 3 of ~ 500 here).

B0 threshold. In EICrecon digitization the energy deposited in each pixel is summed, and only pixels above the threshold become hits (`B0TRK:B0TrackerRawHits:threshold`, default 10 keV).

Purity, cut efficiency. For a selection without truth: purity = signal candidates / selected candidates; efficiency of the cut = signal candidates kept / signal candidates before the cut. Truth is used only to label candidates.

4 First look: B0 hits and the missing tracks

The 100-event samples were first reconstructed with the EICrecon of the eic-shell nightly (`a7bab3f`). The B0 hits are in `B0TrackerHits`. Tracing each hit's MC particle up the parent chain (`b0_hit_origins.C`) shows that in the old geometry 392 hits come from particles whose parent is the Λ (182 π^-, 171 p, and 39 π^+, $K^\pm$, n from Λ interactions in material), and 1086 from deeper

descendants, mostly shower electrons and positrons (818). Yet only 4 B0 tracks were found in 3 of 100 events.

The cause is the truth seeder: `TrackParamTruthInit` only seeded particles produced at $|z| < 200\,\text{mm}$, so daughters of Λs decaying metres downstream had no seed. The four seeded events were exactly the $p\pi^-$ decays at $z < 200\,\text{mm}$. EICrecon#2901 (merged 2026-10-31, in main but not yet in the nightly) adds a dedicated `B0TrackerTruthSeeds` with $|x|, |y| < 240\,\text{mm}$, $z < 6\,\text{m}$, and the perigee at the true production vertex. Table 1 compares the two EICrecon versions on the same simulated events. The hits are identical; the whole gain comes from seeding. The seeder takes every charged particle above 100 MeV in its window, including shower particles, hence ~ 10 seeds per event and the need for truth matching.

Table 1: 100 events per geometry: container EICrecon (a7bab3f) $\rightarrow$ EICrecon main (8b44cae). Events with an entry / total entries (`compare_tracks.C`).

	old geometry	new geometry
B0 simulated hits	77 / 1478	64 / 651
B0 reconstructed hits	76 / 1416	64 / 593
truth seeds	4 / 8 $\rightarrow$ 61 / 650	2 / 4 $\rightarrow$ 59 / 504
B0 tracks	3 / 4 $\rightarrow$ **45 / 77**	1 / 1 $\rightarrow$ **30 / 44**
events with a + and a − track	1 $\rightarrow$ 18	0 $\rightarrow$ 14

All further results use EICrecon main.

5 Geometric acceptance along the Λ direction

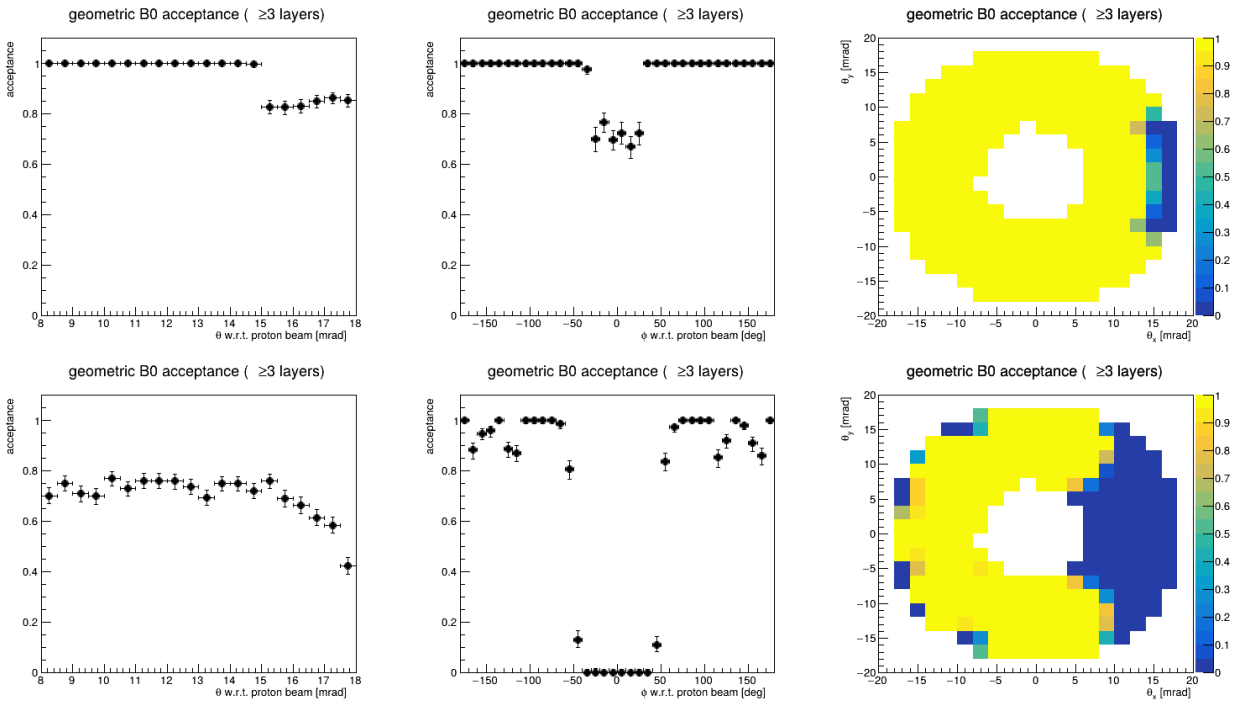


Figure 1: Geometric B0 acceptance (≥ 3 layers) for straight lines from the IP (geantinos) vs θ, ϕ and (θ_x, θ_y) w.r.t. the proton beam. Top: old geometry; bottom: new geometry.

Geantinos have no charge and no interactions; ddsim must be run with `-filter.tracker ""` to keep their (zero-energy) hits. Results (5000 events each):

	old	new
≥ 1 layer	95.8 %	83.5 %
≥ 3 layers	**95.3 %**	**70.1 %**
4 layers	93.2 %	66.9 %

The old geometry covers everything up to 15 mrad and loses only $\theta > 15$ mrad on the side of the electron beam. The new geometry has a gap of about $\pm 45°$ in ϕ around $\phi = 0$ (the electron beam pipe side, $\theta_x \gtrsim +8$ mrad), where nothing is accepted, and extra losses above 16 mrad (Fig. 1).

6 Acceptance for the Λ daughters

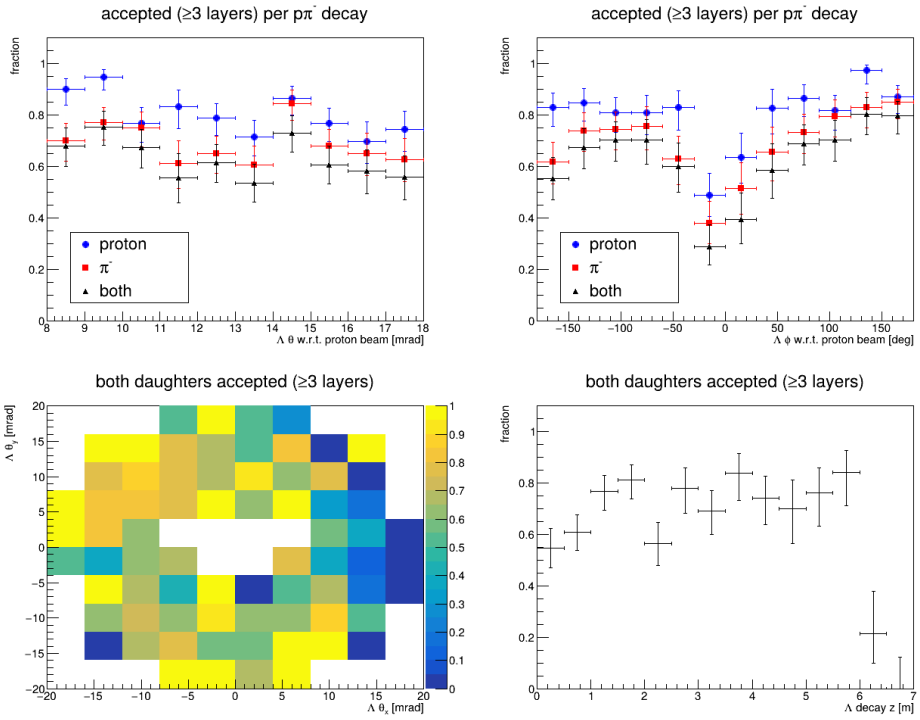
6.1 Decay kinematics

The decay releases only 38 MeV: in the Λ rest frame both daughters have $p^* = 0.1006$ GeV, with $E_p^* = 0.9435$ GeV and $E_\pi^* = 0.1723$ GeV. Boosted to $p_\Lambda = 50$–100 GeV, both move at nearly the Λ velocity and share its momentum roughly as their rest-frame energies:

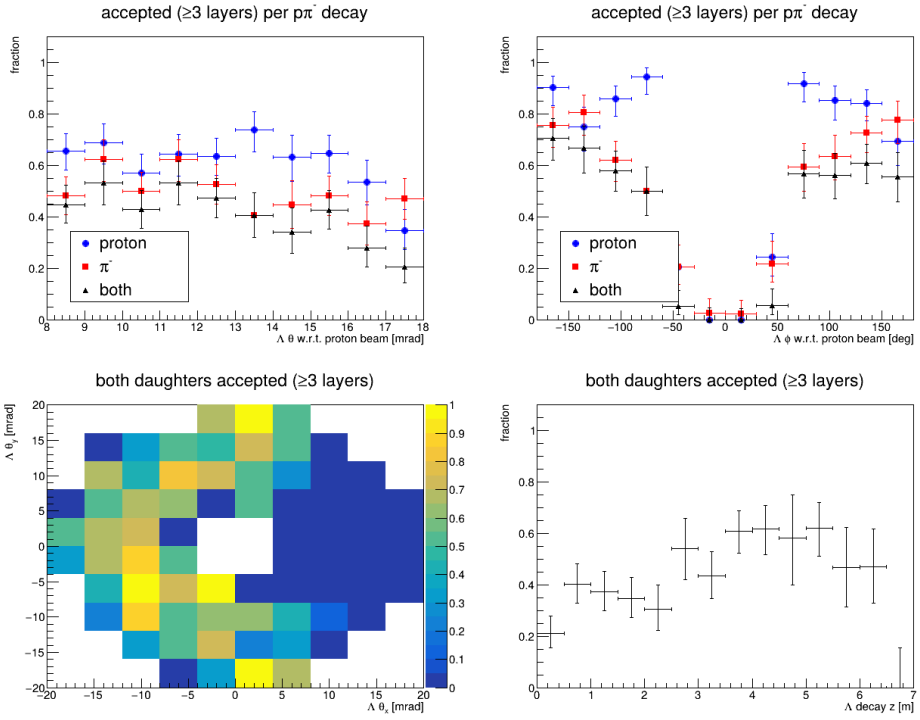
$$\frac{p_p}{p_\Lambda} \approx \frac{E_p^* + p^* \cos\theta^*}{m_\Lambda} \in [0.76, 0.94], \qquad \frac{p_\pi}{p_\Lambda} \approx \frac{E_\pi^* - p^* \cos\theta^*}{m_\Lambda} \in [0.06, 0.25],$$

i.e. protons of 38–94 GeV and π^- of 3–25 GeV. The soft π^- is bent much more by the B0 dipole.

6.2 Results



(a) old geometry



(b) new geometry

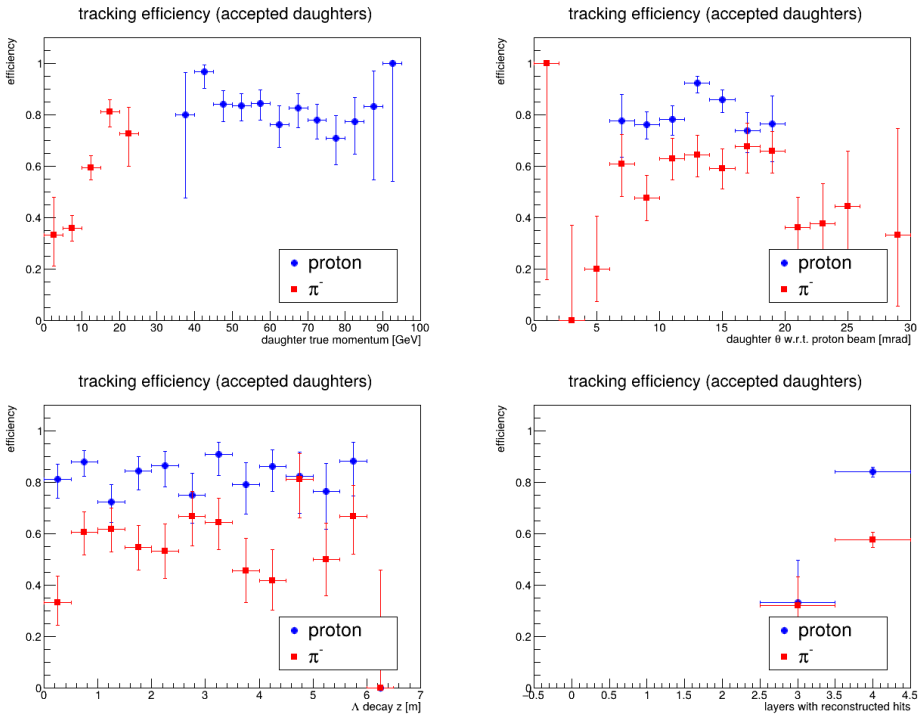
Figure 2: Daughter acceptance per $p\pi^-$ decay (proton blue, π^- red, both black) vs the Λ θ and ϕ, 2D map vs the Λ (θ_x, θ_y) for both daughters, and vs the Λ decay z. 1000 events per geometry.

Table 2: Daughter acceptance per $\Lambda \to p\pi^-$ decay (485–503 decays; statistical error $\approx 2\,\%$).

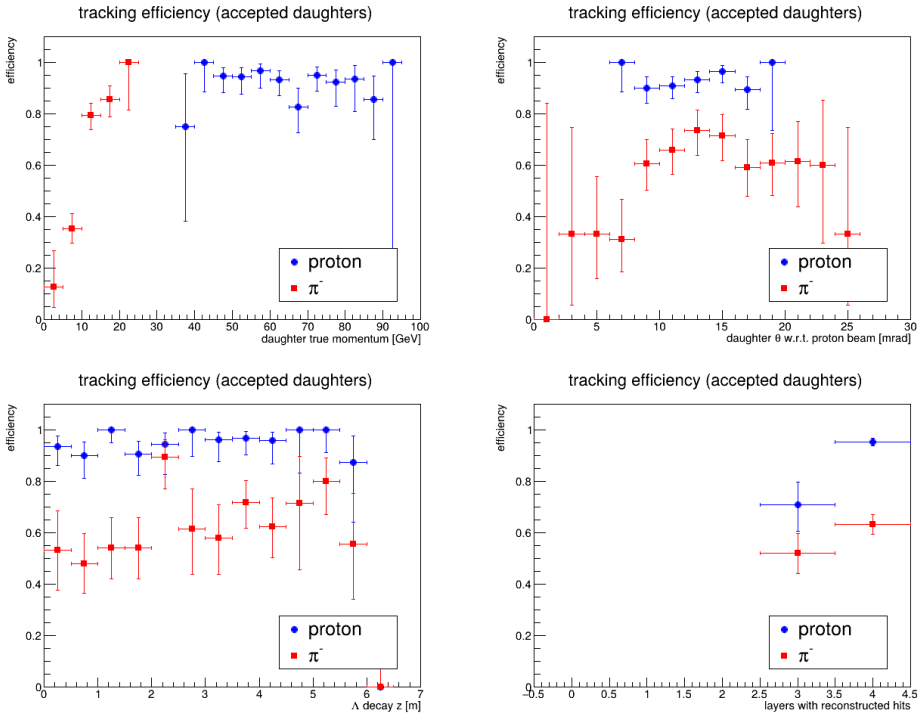
	old geometry	new geometry
proton, ≥ 3 layers	80.3 %	60.8 %
π^-, ≥ 3 layers	69.4 %	49.5 %
both, ≥ 3 layers	**63.2 %**	**41.0 %**
both, ≥ 1 layer	74.6 %	60.0 %
both accepted, per generated Λ	31.8 %	19.9 %

The daughter acceptance is about 30 percentage points below the geometric number: the decay position, the opening angle between the daughters and the bending of the π^- all cost. The ϕ gap of the new geometry is visible as a near-zero acceptance around $\phi = 0$, while the old geometry only has a dip to 30–40 % there. The acceptance rises with the decay distance: early decays give the daughters more distance to spread away from the Λ direction; in the new geometry only $\approx 20\,\%$ of decays near the IP have both daughters accepted. Decays beyond 6 m happen inside the tracker and leave too few layers.

7 Tracking efficiency



(a) old geometry



(b) new geometry, 5 keV

Figure 3: Tracking efficiency for accepted daughters (proton blue, π^- red) vs the daughter's true momentum, its θ w.r.t. the proton beam, the Λ decay z, and the number of layers with reconstructed hits.

Table 3: Accepted daughters passing each step of the reconstruction (`lambda_tracking_efficiency.C`), proton / π^-. Seeds are matched by charge and direction (they carry no truth link).

step	old geometry	new, 10 keV	new, 5 keV
accepted	404 / 349	295 / 240	295 / 240
rec. hits in ≥ 3 layers	100 / 100 %	**93 / 93 %**	100 / 100 %
seed	99 / 99 %	97 / 96 %	97 / 96 %
CKF track	82 / 56 %	86 / 58 %	93 / 61 %
filtered track	82 / 56 %	86 / 58 %	93 / 61 %
reconstructed particle	**82.2 / 55.6 %**	**86.1 / 57.5 %**	**92.9 / 60.8 %**

Where the daughters are lost. Seeds exist for 96–99 % of the accepted daughters; the loss is entirely in the CKF step (seed → track). Track filtering and particle building lose nothing (Table 3).

It is a momentum effect. The π^- efficiency depends strongly on its momentum (Fig. 3, top left): 12–36 % below 10 GeV, 75–90 % at 10–25 GeV, comparable to the proton (80–90 %, flat over 35–95 GeV). The π^- is low on average only because most π^- are soft. Daughters with reconstructed hits in all four layers are reconstructed more often than with three.

Seed momentum smearing ruled out. The truth seeder smears the seed momentum by 10 % (`momentumSmear`) and the seed starts at the decay point, often metres before B0 and the dipole. Rerunning one sample (new geometry, 500 events) with the option `-Ptracking:B0TrackerTruthSeeds:MomentumSmear=0`
(verified: seed q/p then equals the true q/p) gives 71 instead of 70 of 120 accepted π^- reconstructed: no effect. The cause of the CKF loss for soft tracks is not identified yet; candidates are the CKF χ^2/measurement cuts, multiple scattering at low momentum, and the extrapolation of the final track parameters to a perigee at (0,0,0), a known limitation of EICrecon#2901.

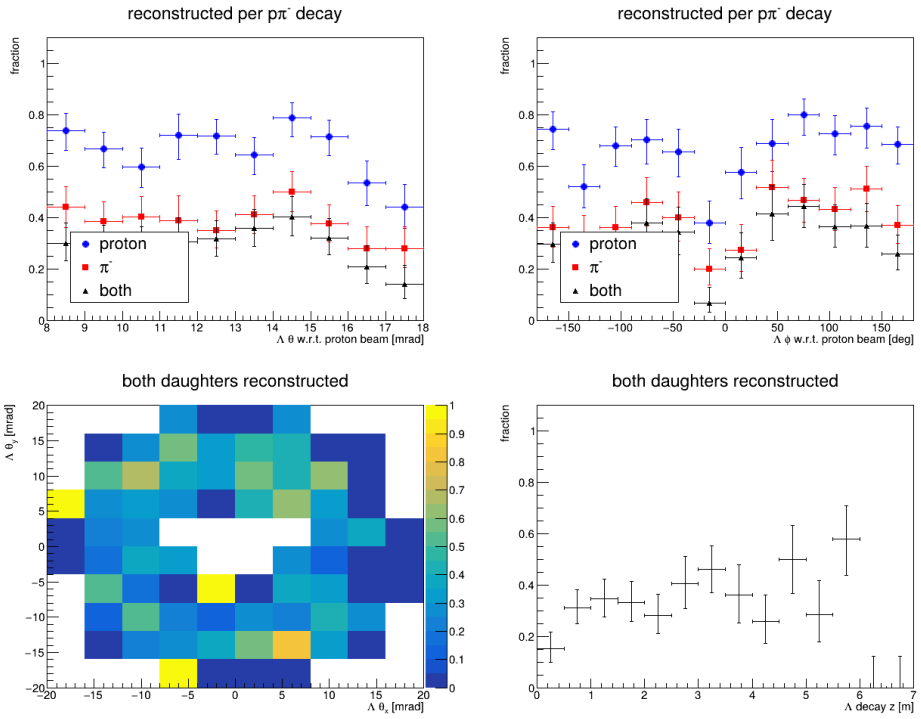
7.1 The B0 threshold

Table 4: Energy deposited per hit by protons, pions and other non-electron particles (500 events per geometry).

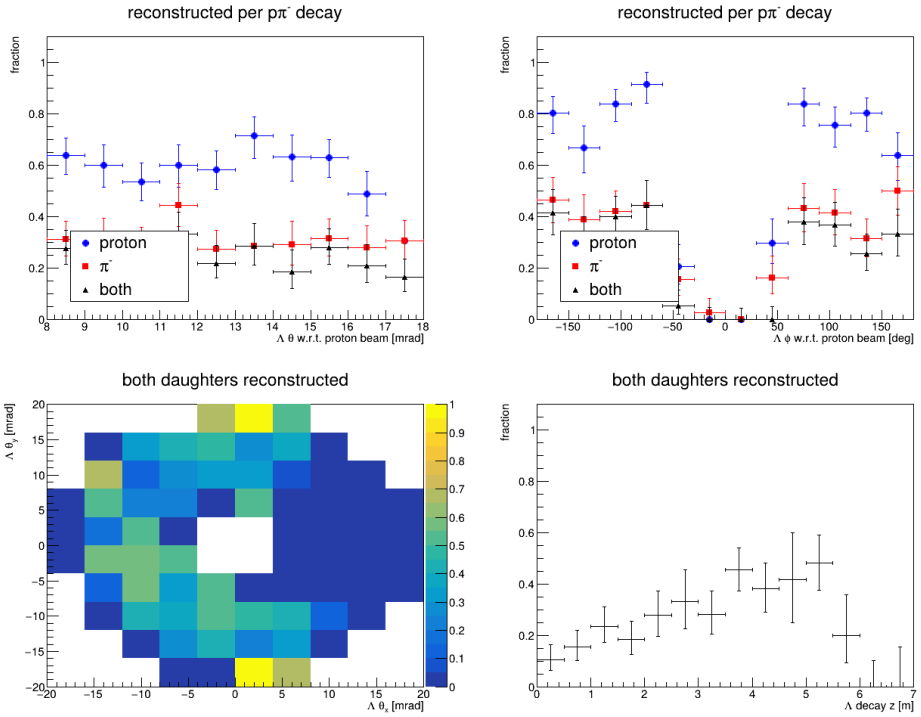
	old geometry	new geometry
sensitive thickness	300 μm	50 μm
most probable deposit	95 keV	13.8 keV
hits below 10 keV	2.6 %	**9.3 %**
hits below 5 keV	1.6 %	0.3 %

A minimum-ionizing particle loses $\approx 0.28\,\text{keV}/\mu\text{m}$ (most probable) in silicon, so the 50 μm sensors give $\approx$ 14 keV, just above the default 10 keV threshold, which cuts the low (Landau) tail: 7 % of the accepted daughters lose hits and fall below three layers of reconstructed hits. With 5 keV (`-PB0TRK:B0TrackerRawHits:threshold=5e-6`) all are recovered and the tracking efficiency of the proton rises from 86 % to 93 %. In the old geometry the threshold makes no difference. The 5 keV setting is not yet the EICrecon default.

8 Acceptance × efficiency



(a) old geometry



(b) new geometry, 5 keV

Figure 4: Acceptance × efficiency per $p\pi^-$ decay: same panels and denominator as Fig. 2, numerator = reconstructed daughters.

Table 5: Acceptance, tracking efficiency and acceptance × efficiency (`lambda_acceptance_x_efficiency.C`), per $p\pi^-$ decay: proton / π^- / both.

	old geometry	new, 10 keV	new, 5 keV
acceptance A	80 / 69 / 63 %	61 / 49 / 41 %	61 / 49 / 41 %
tracking efficiency ε	82 / 56 / 47 %	86 / 58 / 50 %	93 / 61 / 59 %
$A \times \varepsilon$	66 / 39 / **30 %**	53 / 29 / **21 %**	57 / 31 / **25 %**
both reconstructed, per generated Λ	**15.0 %**	**10.1 %**	**12.0 %**

The direct ratio agrees with the product $A \cdot \varepsilon$ to within 0.6 %; at most three daughters per sample are reconstructed without being accepted. The new geometry at 5 keV recovers part of the loss of the new geometry, but stays below the old geometry because of its smaller acceptance.

9 The Λ mass

9.1 Truth-matched pairs

For each $p\pi^-$ decay with both daughters reconstructed, $M(p\pi^-)$ is computed from the reconstructed momenta with PDG masses (`lambda_mass_reco.C`).

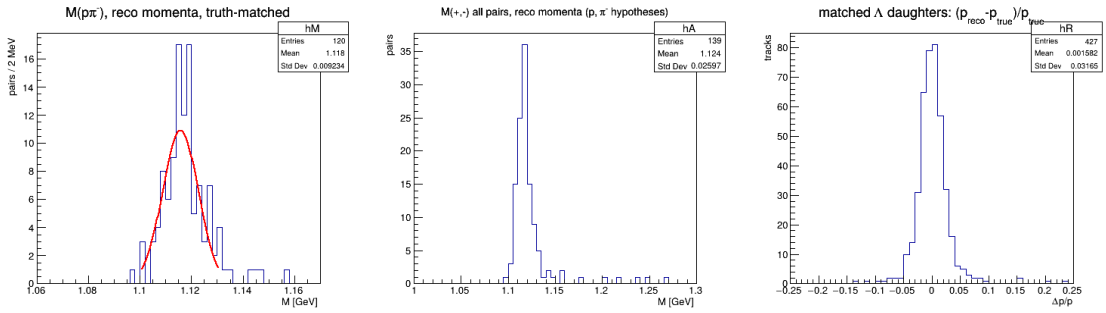


Figure 5: New geometry, 5 keV: $M(p\pi^-)$ of truth-matched pairs with a Gaussian fit of the core (left), all $(+,-)$ pairs without selection (middle), and the momentum resolution of the matched daughters (right).

	old geometry	new, 10 keV	new, 5 keV
truth-matched pairs (of 1000 Λ)	150	101	120
Gaussian mean [GeV]	1.1177 ± 0.0007	1.1160 ± 0.0010	1.1159 ± 0.0008
Gaussian σ [MeV]	6.3 ± 0.6	7.4 ± 1.1	6.9 ± 1.1
$\Delta p/p$ RMS of matched daughters	4.4 %	3.2 %	3.2 %

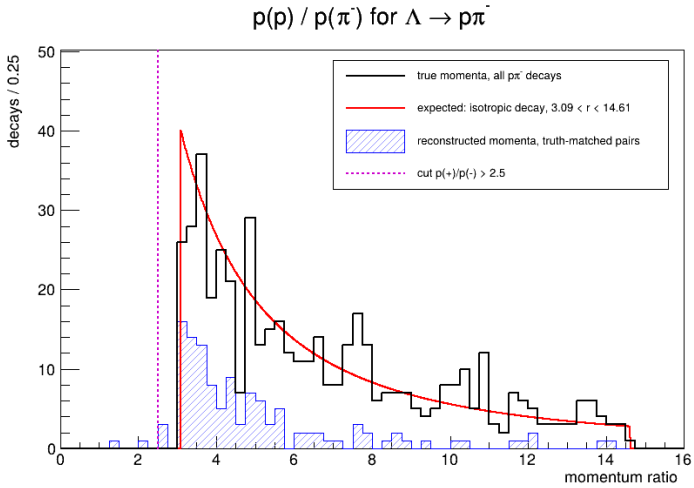
The new geometry peaks at the PDG mass (1.1157 GeV). The old geometry is ≈ 2 MeV high ($\approx 3\sigma$ of the fit error), not understood. The width is 6–7 MeV in all samples; the better momentum resolution of the new geometry does not yet show at this statistics.

9.2 Without truth in the selection

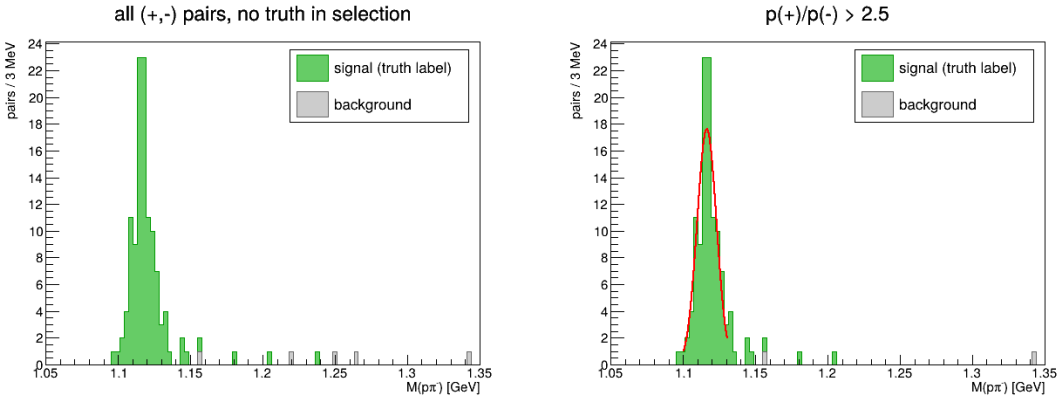
Candidates are all $(+,-)$ pairs of reconstructed particles in an event, with the $+$ track as the proton and the $-$ track as the π^-. From the kinematics above, γ cancels in the momentum ratio:

$$r = \frac{p_p}{p_\pi} = \frac{E_p^* + p^* \cos\theta^*}{E_\pi^* - p^* \cos\theta^*} \in \left[\frac{0.8429}{0.2729}, \frac{1.0441}{0.0717}\right] = [3.09, 14.6]$$

for any Λ momentum, and for an isotropic decay ($\cos\theta^*$ uniform) its distribution is $f(r) = (E_p^* + E_\pi^*)/[2p^*(1+r)^2]$. The true ratio indeed spans 3.09–14.58 and follows $f(r)$ (Fig. 6a). The selection $p(+)/p(-) > 2.5$ leaves room for mismeasurement; it loses $\approx 2\,\%$ of the signal (badly measured tracks).



(a) momentum ratio



(b) mass without truth

Figure 6: New geometry, 5 keV. (a) True $p(p)/p(\pi^-)$ (black), the expectation for an isotropic decay (red), and the reconstructed ratio of truth-matched pairs (blue); dashed: the cut. (b) $M(p\pi^-)$ of all $(+,-)$ pairs (left) and after the cut (right); signal (green, truth label) stacked on background (grey).

	old geometry	new, 10 keV	new, 5 keV
all $(+,-)$ pairs / purity	201 / 75 %	116 / 87 %	139 / 86 %
after $p(+)/p(-) > 2.5$ / purity	155 / **95 %**	103 / **96 %**	122 / **97 %**
signal kept by the cut	98 %	98 %	98 %
±20 MeV window: candidates / signal	144 / 142	92 / 92	110 / 110
Gaussian mean [GeV] / σ [MeV]	1.1174 / 6.6	1.1164 / 6.4	1.1166 / 6.7

The background (shower tracks, mainly in the old geometry) lies far from the peak. Note that the tracks are still truth-seeded and the events contain a single Λ and nothing else.

10 Summary and outlook

- B0 tracks for displaced Λ daughters require the B0 truth seeding of EICrecon#2901: $4 \to 77$ tracks per 100 Λ (old geometry).
- The realistic B0 geometry (eic/epic#1194) has a $\pm 45°$ gap in ϕ on the electron-beam side: geometric acceptance 95 % $\to$ 70 %, daughter acceptance (both) 63 % $\to$ 41 %.
- Its 50 μm sensors need a 5 keV threshold (10 keV loses 7 % of the accepted daughters).
- Tracking efficiency for accepted daughters: proton 82–93 %, π^- 56–61 %. Soft tracks ($<$ 10 GeV) are lost in the CKF; not due to the seed momentum smearing.
- Both daughters reconstructed: 15 % (old), 10 % (new, 10 keV), 12 % (new, 5 keV) of the generated Λs.
- $M(p\pi^-)$ at the PDG value with $\sigma \approx$ 6–7 MeV; a single kinematic cut gives $\approx$ 97 % purity without truth.

Next steps: understand the CKF loss of soft tracks (debug output on a few events, CKF cuts); realistic B0 seeding (EICrecon#3028) instead of truth seeding; cross-check with the ChargedLambdaReconstruction of EICrecon#2926; the 2 MeV mass shift in the old geometry; events with other particles and beam background, where the decay vertex will be needed.

A Reproducing the results

- Builds: `scripts/build/build_epic.sh`, `scripts/build/build_eicrecon.sh`.
- Samples: `scripts/production/lambda_sample.sh <GEO> <NAME> <SEED> <N>` (simulation, reconstruction at 10 and 5 keV, copy to EOS) and `scripts/production/geantino_sample.sh`.
- Analysis: `scripts/make_results.sh` runs every macro and writes `results/`.

B Pitfalls met

- `eic-shell - bash -c ...` fails ("Illegal option –"); scripts are piped into eic-shell (`scripts/in_eic_shell.sh`), with `set -` before sourcing `thisepic.sh`.
- EOS is not visible inside eic-shell: jobs write to the node's `/tmp`, then copy.
- Parallel jobs in one folder can read each other's half-written `calibrations/` files.
- ddsim drops tracker hits below 1 keV by default (all geantino hits).
- EICrecon parameters need the plugin prefix (`tracking:`, `BOTRK:`); an unknown key stops eicrecon at start-up.